



Bapuji Educational Association ®
Bapuji Institute of Engineering and Technology, Davangere-577004
An Autonomous Institute affiliated to VTU
Department of Mathematics

Course Title	Mathematics for Computer Science and Engineering	Course Code	BCSASC301
Course Instructor	Dr. VK, Dr. PRJ, Dr. HBS, GSP, Dr. ABV, RMT, PGH, KPM, ARS, PKN, APC, Dr. JPK, Dr. MBR	Semester Branch	III CS, AIML, IS, CSDS, CSBS & CSDE
I. A. Test No.	I	Max. Marks	30
Date	14.11.2025	Time	9.30 – 10.30 AM

Course Outcome Statements: After the successful completion of the course, the students will be able to

CO1 Explain the basic concepts of probability distributions, Markov chain, statistical inference and ANOVA

CO2 Apply the techniques of suitable probability distribution models, the notion of a discrete-time Markov chain to solve the engineering problems

CO3 Apply the Concepts of suitable statistical methodology to meet specified requirements of various engineering problems

CO4 Analyze the data and interpret the results using statistical inferences

CO5 Analyze the variations using ANOVA technique

Note: Answer any one full question from each part.

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Q. No	Question		Marks	RBT level	CO																	
PART A																						
1	a	Derive an expression for mean, variance and standard deviation of Binomial distribution.	8	L2	1																	
	b	A random variable X has the following probability function for various values of x	7	L3	2																	
		<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>P(x)</td><td>0</td><td>k</td><td>2k</td><td>2k</td><td>3k</td><td>k²</td><td>2 k²</td><td>7 k²+k</td></tr></table>				x	0	1	2	3	4	5	6	7	P(x)	0	k	2k	2k	3k	k ²	2 k ²
x		0				1	2	3	4	5	6	7										
P(x)	0	k	2k	2k	3k	k ²	2 k ²	7 k ² +k														
Determine i) the value of k ii) $P(x < 6)$ iii) $P(0 < x < 5)$																						
OR																						
2	a	Derive an expression for mean, variance and standard deviation of Poisson distribution.	8	L2	1																	
	b	In a certain town the duration of shower is exponentially distributed with mean 5 minutes. Determine the probability that the shower will last for, (i)10 minutes or more (ii) less than 10 minutes (iii) between 10 and 12 minutes	7	L3	2																	
PART B																						
3	a	A coin is tossed 1000 times and head turns up 540 times. Investigate whether the coin is unbiased at 1% level of significance.	7	L3	4																	
	b	A survey was conducted in a slum locality of 2000 families by selecting a sample of size 800 families. It was revealed that 180 families were illiterates. Determine the probable limits of the illiterate families in the population of 2000.	8	L3	4																	
OR																						
4	a	The mean life time of a sample of 100 fluorescent tube lights manufactured by a company is found to be 1570 hrs with a standard deviation of 120 hrs. Investigate whether the mean life-time of the lights produced by the company is 1600 hrs at 0.01 level of significance.	7	L3	4																	
	b	In two large populations there are 30% and 25% respectively of fair-haired people. Investigate whether this difference is likely to be hidden in samples of 1200 and 900 respectively from the two populations at 5% significance level.	8	L3	4																	

RBT(Revised Bloom's Taxonomy) Levels: Cognitive Domain

L1: Remembering L2: Understanding L3: Applying L4: Analyzing L5: Evaluating L6: Creating

Course Coordinator

Coordinator: DAAC

Program Coordinator

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I. A. Test No.	II	Max. Marks	30
Date	08.01.2026	Time	9.30 – 10. AM

Course Outcome Statements: After the successful completion of the course, the students will be able to	
CO1	Explain the basic concepts of probability distributions, Markov chain, statistical inference and ANOVA
CO2	Apply the techniques of suitable probability distribution models, the notion of a discrete-time Markov chain to solve the engineering problems
CO3	Apply the Concepts of suitable statistical methodology to meet specified requirements of various engineering problems
CO4	Analyze the data and interpret the results using statistical inferences
CO5	Analyze the variations using ANOVA technique

Note: Answer any one full question from each part.

Q. No	Question	Marks	RBT level	CO												
PART A																
1	The joint probability distribution of discrete random variables X & Y is as follows: <table><tr><td>$\begin{matrix} & Y \\ X \backslash & \end{matrix}$</td><td>-4</td><td>2</td><td>7</td></tr><tr><td>1</td><td>1/8</td><td>1/4</td><td>1/8</td></tr><tr><td>5</td><td>1/4</td><td>1/8</td><td>1/8</td></tr></table> Compute i) E(X) and E(Y) ii) E(XY) iii) σ_X and σ_Y iv) $\rho(X, Y)$	$\begin{matrix} & Y \\ X \backslash & \end{matrix}$	-4	2	7	1	1/8	1/4	1/8	5	1/4	1/8	1/8	8	L3	2
	$\begin{matrix} & Y \\ X \backslash & \end{matrix}$	-4	2	7												
1	1/8	1/4	1/8													
5	1/4	1/8	1/8													
b	Show that the matrix $A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 2 & 4 & 4 \\ 0 & 1 & 0 \end{bmatrix}$ is a regular stochastic matrix	7	L3	2												
OR																
2	The joint probability distribution of two discrete random variables X and Y is $f(x, y) = k(2x + y)$, where x and y are integers such that $0 \leq x \leq 2, 0 \leq y \leq 3$ i) Determine the value of constant k. ii) Show that the random variables X and Y are dependent. iii) Determine $P(X \geq 1, Y \leq 2)$	8	L3	2												
	The students study habits are as follows; If he studies on one night, he is 60% sure not to study on next night. On the other hand, if he does not study on to night, he is 80% sure to study next night. Write the transition probability matrix for his chain of study. In the long run, investigate how often does he study? Suppose he studies on Monday night, determine the probability that he does not study on Friday night?	7	L3	2												

PART B

PART B																														
3	a	Explain One -way ANOVA.						3	L2	1																				
	b	Three types of fertilizers are used on three groups of plants for 5 weeks. Examine whether there is a difference in the mean growth of each group. Using the data given below apply a one-way ANOVA test at 0.05 significant level. (Given that $F_{2,15} = 3.68$) <table><tr><td>Fertilizer-1</td><td>6</td><td>8</td><td>4</td><td>5</td><td>3</td><td>4</td></tr><tr><td>Fertilizer-2</td><td>8</td><td>12</td><td>9</td><td>11</td><td>6</td><td>8</td></tr><tr><td>Fertilizer-3</td><td>13</td><td>9</td><td>11</td><td>8</td><td>7</td><td>12</td></tr></table>						Fertilizer-1	6	8	4	5	3	4	Fertilizer-2	8	12	9	11	6	8	Fertilizer-3	13	9	11	8	7	12	12	L4
Fertilizer-1	6	8	4	5	3	4																								
Fertilizer-2	8	12	9	11	6	8																								
Fertilizer-3	13	9	11	8	7	12																								

OR

4	a	Explain Two -way ANOVA.	3	L2	1																																																	
	b	Latin-square design experiment was conducted in respect of five fertilizers which were used on plots of different fertility. <table><tr><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td></tr><tr><td>16</td><td>10</td><td>11</td><td>9</td><td>9</td></tr><tr><td>E</td><td>C</td><td>A</td><td>B</td><td>D</td></tr><tr><td>10</td><td>9</td><td>14</td><td>12</td><td>11</td></tr><tr><td>B</td><td>D</td><td>E</td><td>C</td><td>A</td></tr><tr><td>15</td><td>8</td><td>8</td><td>10</td><td>18</td></tr><tr><td>D</td><td>E</td><td>B</td><td>A</td><td>C</td></tr><tr><td>12</td><td>6</td><td>13</td><td>13</td><td>12</td></tr><tr><td>C</td><td>A</td><td>D</td><td>E</td><td>B</td></tr><tr><td>13</td><td>11</td><td>10</td><td>7</td><td>14</td></tr></table> Examine whether there is a significance difference between rows and columns at 5% significance level (Given that $F_{4,12} = 3.26$)	A	B	C	D	E	16	10	11	9	9	E	C	A	B	D	10	9	14	12	11	B	D	E	C	A	15	8	8	10	18	D	E	B	A	C	12	6	13	13	12	C	A	D	E	B	13	11	10	7	14	12	L4
A	B	C	D	E																																																		
16	10	11	9	9																																																		
E	C	A	B	D																																																		
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15	8	8	10	18																																																		
D	E	B	A	C																																																		
12	6	13	13	12																																																		
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L1: Remembering	L2: Understanding	L3: Applying	L4: Analyzing	L5: Evaluating	L6: Creating
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Santosh
29/12/2025
Course Coordinator

A. D.
29/12/2025
Coordinator: DAAC

NB
29/12/25
Program Coordinator